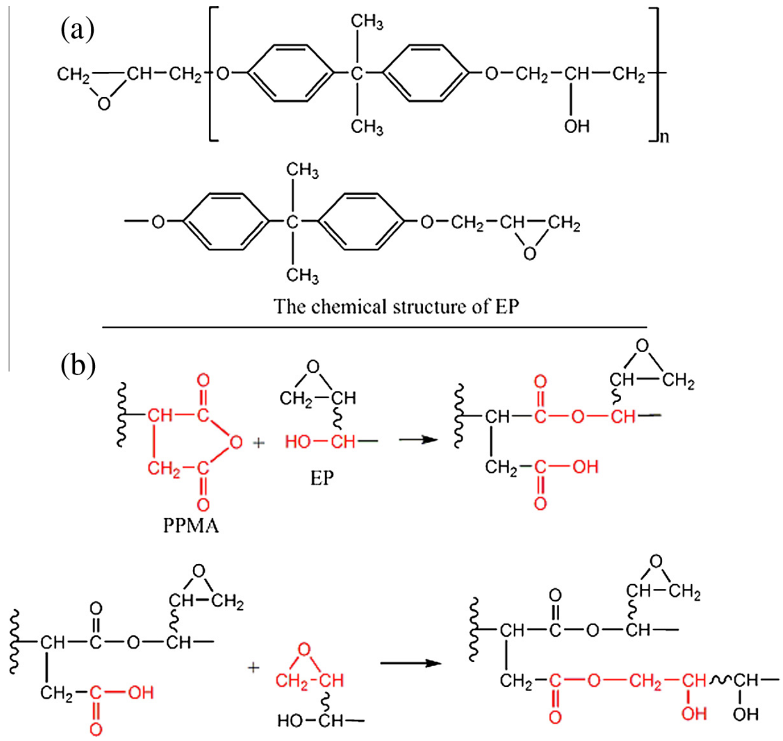


Fig. 5. Chemical structure of EP (a) and possible reaction between PPMA and EP (b).

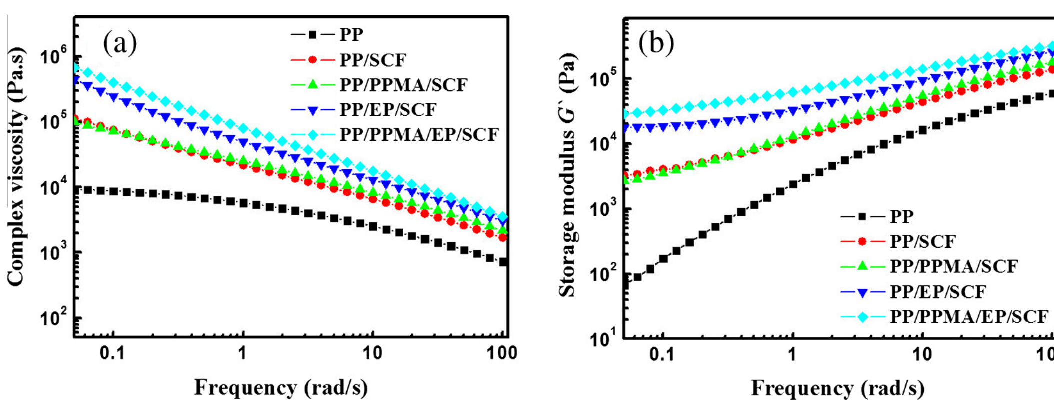


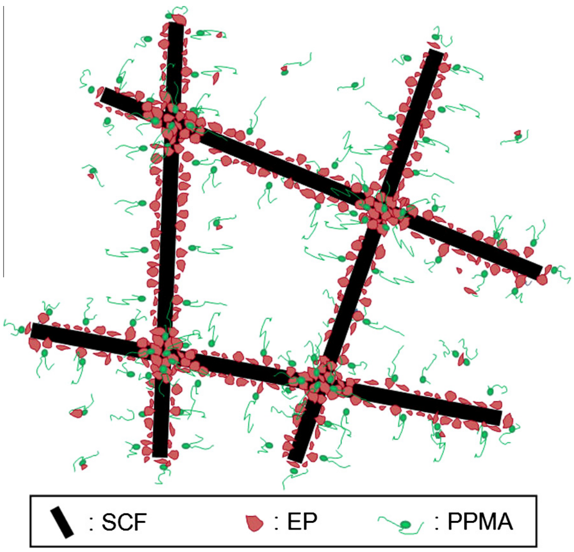
Fig. 6. Complex viscosity (a) and storage modulus (b) of PP and PP composites.

in the PP/PPMA/EP/SCF composites improved not only tensile and flexural properties, but also impact properties.

According to the above results, compared to other composites (PP/SCF, PP/PPMA/SCF and PP/EP/SCF), there are two possible reasons for improved mechanical properties of PP/SCF composites after adding PPMA and EP at the same time. As shown in Scheme 1, four components show the following distribution: PP acts as a matrix (white substrate), SCF acts as reinforced fillers. PPMA and EP consists of interlayer between PP matrix and SCF fillers. In order to form a stable microstructure of the composites, SCF should be wrapped by EP component, and PPMA exists in between EP and PP. This distribution is favorable to reduce free energy in the composites. Firstly, the addition of both PPMA and EP would greatly increase the interfacial interaction between SCF fillers and PP matrix, which can improve stress transfer efficiency from the matrix to the filler. Secondly, the linking sites (or crossover sites) between different SCFs were locked by crosslinked EP (Fig. 3d), as a result, the network structure of SCFs with strong interaction was formed, which could promote the capacity supporting outside loading. The above factors were attributed to the enhanced mechanical properties of the composites together. Comparatively, adding PPMA or EP into PP/SCF composites alone, the interfacial interaction between PP and SCF (or EP layer on the surface of SCF) and stress transfer efficiency between SCFs was not enough.

3.4. Optimizing the composition of PP/PPMA/EP/SCF composites

In order to further improve the mechanical properties of PP/PPMA/EP/SCF composites, we optimized the composition of PP/PPMA/EP/SCF composites. Fig. 7 shows the tensile, flexural and



Scheme 1. Schematic drawing of PP/PPMA/EP/SCF composites (PP as the matrix).

impact properties of the composites as a function of EP content. When the content of EP increased, the mechanical properties (tensile strength, Young modulus, flexural strength, flexural modulus and impact strength) of PP/PPMA/EP/SCF composites first increased and then decreased. The tensile strength and modulus reached the maximum at the content of 7.5 wt% EP. The flexural strength, flexural modulus and impact strength reached the maximum at the content of 5 wt% EP.

Fig. 8 shows the effect of PPMA loading on the tensile, flexural and impact properties of the composites. When the content of PPMA was low than 8 wt%, the tensile strength, Young modulus, flexural strength, flexural modulus and impact strength were increased. The mechanical properties of PP/PPMA/EP/SCF composites reached a maximum in the case containing 8 wt% of PPMA. The improvement in the mechanical properties was ascribed to the enhancing interfacial adhesion between SCF (or EP) and PP matrix. However, the PPMA has poor mechanical properties due to its low molecular weight. So the mechanical properties were decreased with the further increase of PPMA content, when the content of PPMA was over 8 wt%.

As mention-above, there are two aims for adding both PPMA and EP in PP/SCF composites. On one hand, the added PPMA with low molecular weight acts as a compatibilizer, which is expected to locate in the interfacial area to strengthen the interfacial adhesion between PP and SCF. Owing to low molecular weight of PPMA, the presence of more PPMA in PP matrix except the part located at the interfacial area has an adverse effect on the mechanical properties. On the other hand, the addition of EP can help the formation of SCF network structure by locking crossover sites between SCFs (Scheme 1). However, the addition of excess EP will result in the formation of EP dispersed phase directly contacting with PP matrix due to no enough amount of PPMA as compatibilizer. This also will lead to the reduction of mechanical properties of PP/PPMA/EP/SCF composites. As shown in Fig. 6a, the addition of EP increased the melt viscosity of the composites due to curing reaction of EP with imidazole. As a result, the length of SCF in the composites decreased with the increase in EP loading (Fig. 9). The number average length of SCF decreased from 0.27 to 0.18 mm and the

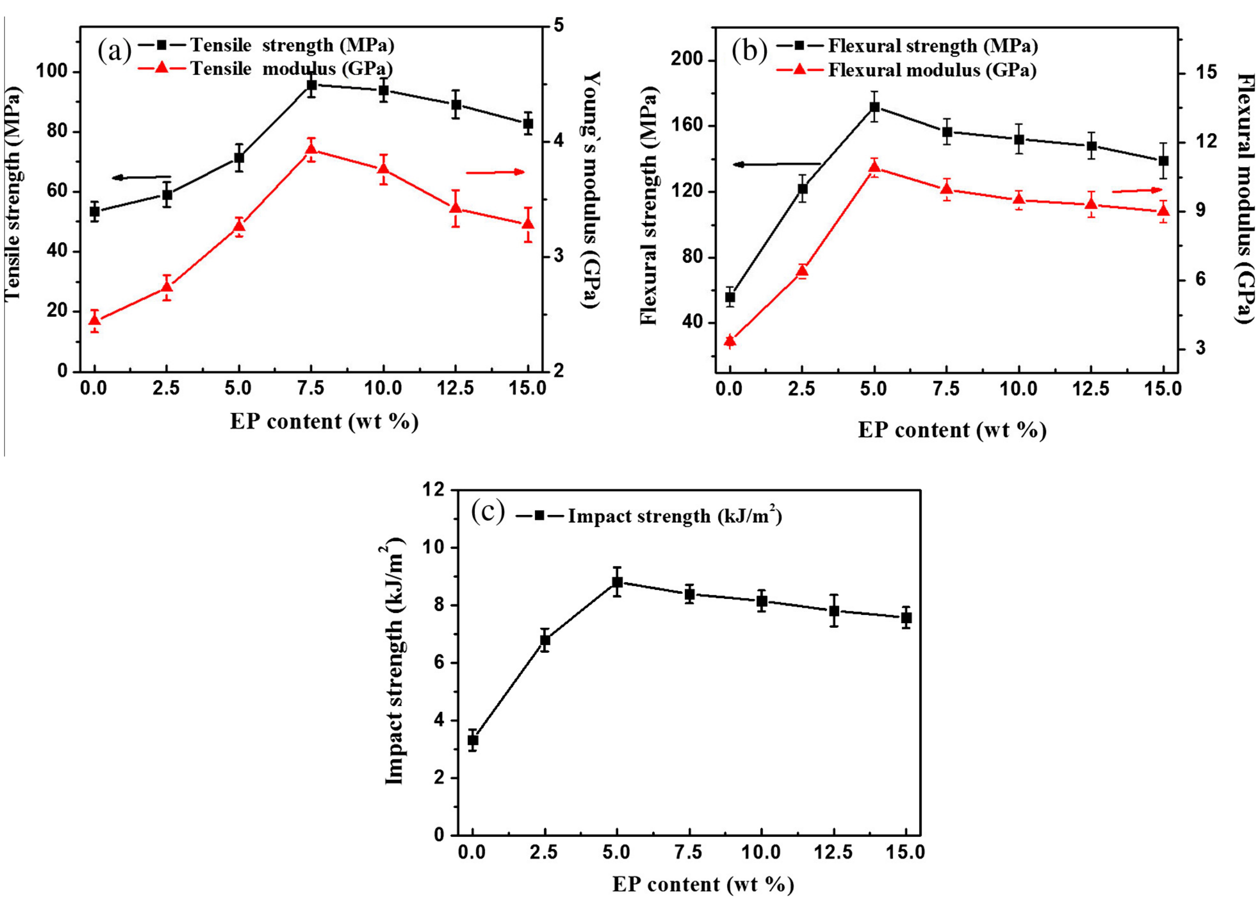


Fig. 7. Mechanical properties of PP/PPMA/EP/SCF composites with different EP content (PPMA: 7.5 wt%; SCF: 26.4 wt%; PP + EP = 66.1 wt%).

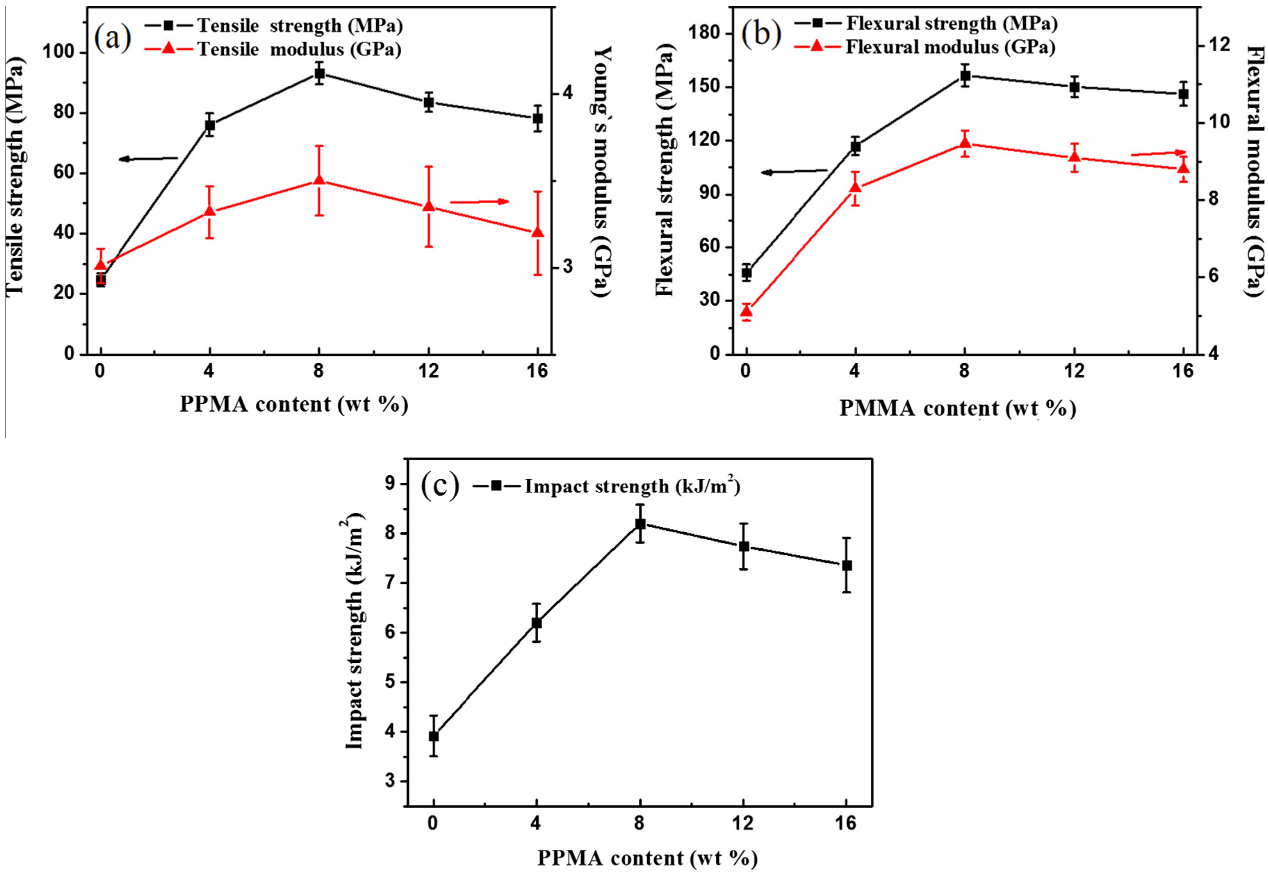


Fig. 8. Mechanical properties of PP/PPMA/EP/SCF composites with different PPMA content (EP: 11.6 wt%; SCF: 26.4 wt%; PP + PPMA = 62 wt%).

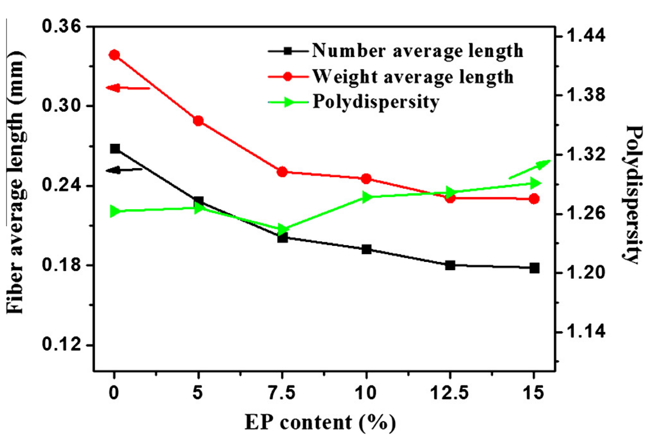


Fig. 9. The influence of EP content on the carbon fiber length in PP/PPMA/EP/SCF composites (PPMA: 7.5 wt%; SCF: 26.4 wt%; PP + EP = 66.1 wt%).

weight average length decreased from 0.34 to 0.23 mm when the EP content increased. The polydispersity index of the fiber length was 1.26, which means a narrow fiber length distribution. Thus the contents of PPMA and EP could be not too high in order to prepare high-performance PP/SCF composites.

4. Conclusions

Epoxy resin was successfully introduced into the PP/PPMA/SCF composite to significantly improve the mechanical properties. SEM, FTIR and rheology measurements confirmed that the improvement of mechanical properties of PP/PPMA/EP/SCF composite was mainly attributed to the synergistic effect of PPMA and EP. The EP formed a protective layer on the carbon fiber surface and locked crossover sites of adjacent carbon fibers to form a strong network structure, and the PPMA reacted with EP protective layer and improved the interfacial interaction between matrix and fillers. Thus the addition of both PPMA and EP would greatly improve stress transfer efficiency from the matrix to the filler. The formation of SCF network structure could promote the capacity supporting outside loading. Therefore the mechanical properties of PP/PPMA/EP/SCF composites were significantly improved compared to the addition of PPMA or EP alone. It is believed that the results in this work will shed a light to prepare high-performance polymer/SCF composites.

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